SRM
INSTITUTE OF SCIENCE & TECHNOLOGY

Faculty of Engineering and Technology
Project Work/Internship
Department of Computer Science and Engineering

Project Batch ID
BT27AIMLC019

Degree/ program	B. Tech CSE	**Specialization**	AI & ML
Academic Year	2024-2025 (Odd)	**Semester**	5th
Course Code & Course Title:	Research Project		
Working Title of the Project:	Swar Manovigyan: Music and Mental Health		
Project Site / Location	https://github.com/Adit-Jain-srm/Swar_Manovigyan_ML		
Name and address of the company (Applicable for projects with industry or industry support):	NA		

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Panel Approval

Approved / Not Approved	
Remarks by Panel Member	
Panelist Name and Signature with Date	

Project Coordinator

HOD CSE

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Student Log Book – Meeting with Guide

S.No	Date of Meeting	Activities or Discussion of the project	Supervisor Comments	Sign of the Supervisor with Date
1	09/10/2025	Discussion About Abstract & Title	OK	9/10/25
2				
3				
4				

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Project Details

1. Research Domain:

AI & Machine Learning, Data Science & Analytics

2. Abstract of the Research Project (~200 words)

Music therapy has emerged as a promising approach for mental health treatment, yet current music recommendation systems lack the emotional granularity needed for therapeutic applications. This paper presents Swar_Manovigyan_ML, **an emotion-aware music therapy system that predicts continuous arousal-valence (A/V)** coordinates from audio features using a Bidirectional Long Short-Term Memory (BiLSTM) neural network.

The system employs the dimensional emotion model to map musical content to a two-dimensional psychological space, where arousal represents energy levels (calm to excited) and valence represents emotional tone (negative to positive). Our approach extracts mel-spectrogram features for temporal modeling and implements a deep learning architecture with **bidirectional LSTM layers (96 and 48 units) followed by dense layers (128 and 64 units) to predict continuous A/V values in the range [0,1]**. The system provides personalized music recommendations by mapping predicted A/V coordinates to four emotion quadrants (sad/depressed, calm/peaceful, angry/stressed, happy/excited) with genre-specific therapeutic suggestions.

A user-friendly Streamlit interface enables real-time audio analysis and visualization of emotional content, making the system accessible for therapeutic applications. The model achieves competitive performance using Mean Squared Error (MSE) loss and Mean Absolute Error (MAE) metrics, with early stopping and learning rate reduction callbacks for optimization.

3. Objective:

The objective of this project is to design and develop an **emotion-aware music therapy system** that can predict a listener's emotional state from musical audio using a **deep learning–based arousal–valence (A/V) regression model**.

The system aims to bridge the gap between traditional music emotion recognition and therapeutic applications by:

- Modeling emotional states on a **continuous arousal–valence scale** instead of discrete mood labels.
- Extracting **mel-spectrogram** and **high-level audio features** that capture the emotional characteristics of music.
- Developing a **BiLSTM-based neural architecture** capable of learning temporal emotional patterns from audio sequences.
- Providing **personalized therapeutic music recommendations** based on predicted emotion quadrants.
- Offering a **user-friendly interface** for real-time emotion analysis and music suggestion.

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4. Expected Outcomes / Deliverables:

The project will deliver the following key outcomes:

1. Emotion Prediction Model

- A trained **BiLSTM regression model** for predicting continuous arousal and valence values in the range [0,1].
- Performance evaluation using **MSE** and **MAE** metrics on a test dataset.
- Optimized training using early stopping, learning-rate scheduling, and standardized feature sequences.

2. Audio Feature Extraction Pipeline

- A complete feature engineering module including:
    - Mel-spectrogram extraction
    - Spotify-style audio features (energy, danceability, acousticness, tempo, etc.)
    - Computed emotional features: **arousal** and **enhanced valence**

3. Emotion Quadrant Classification

- Conversion of continuous A/V predictions into four therapeutic quadrants:
    - Sad / Depressed
    - Calm / Peaceful
    - Angry / Stressed
    - Happy / Excited

4. Therapeutic Recommendation Engine

- A rule-based module mapping emotion quadrants to **specific genres and mood-appropriate music recommendations** for mental well-being.

5. Streamlit-based Interactive Interface

- Real-time audio upload and emotional analysis
- Plotly-based A/V visualization
- Interactive quadrant explanation and recommendations
- Option for manual inputs and feature-based prediction modes

6. End-to-End Deployment-Ready System

- Fully functional codebase supporting:
    - Training
    - Inference
    - Audio preprocessing
    - Streamlit-based UI
    - Model loading and real-time predictions

7. Research Documentation

- Complete technical report including:
    - Literature review
    - Methodology
    - Experiments and results
    - Limitations and future work